

ABDELRAHMAN ALORDY

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RESEARCH INTERESTS

- Robotic Manipulation in Unstructured Environments
- Deep Learning for Perception & Control
- Multi-Modal Learning & Optimization
- Sim-to-Real Transfer & Domain Adaptation
- Human-Robot Interaction
- Reinforcement Learning & Multi-Agent Systems

EDUCATION

B.Sc. in Computers and Information (Software Engineering)

Cairo University

Oct. 2020 – May. 2025

Giza, Egypt

- **GPA:** 3.4/4.0
- **Coursework:** Artificial Intelligence, Machine Learning, Algorithms, Data Structures, Computer Architecture, Databases

Visiting Student (Study Abroad Semester)

University of Louisville

Aug. 2023 – Dec. 2023

Louisville, KY, USA

- **Coursework:** Embedded Systems, Data Analytics, Data Analysis and Decision Making

RESEARCH & INTERNSHIPS

AI & Robotics Research Intern

Singapore Institute of Manufacturing Technology (SIMTech), A*STAR

Sep. 2024 – Dec. 2024

Singapore

- Collaborated with PhD researchers to optimize a deep learning model for a UR10e robotic grasping system, using PyTorch, OpenCV, and AnyGrasp, applying multimodal learning to integrate visual and kinematic data for improved performance.
- Developed a real-time microservices-based Python application integrating the RealSense camera, grasp detection model, Robotiq gripper, and robotic control via ROS2 and RTDE, enhancing system efficiency and responsiveness.

Machine Learning Research Intern

Meshcheryakov Laboratory of Information Technologies (MLIT), JINR

Aug. 2024 – Sep. 2024

Moscow, Russia

- Collaborated with physicists to improve primary vertex detection using machine learning techniques (e.g., Scikit-learn, TensorFlow), achieving a 9% accuracy improvement over existing algorithms.
- Fine-tuned an event reconstruction algorithm in C++ for the Multi-Purpose Detector, optimizing data flow for detection.

Software Engineer Intern

Halliburton

Jul. 2023 – Aug. 2023

Cairo, Egypt

- Developed backend services in Java for report uploads, task management, and tracking well/reservoir status for field engineers, using Spring Boot, Docker, and PostgreSQL for efficient data handling.

SELECTED PROJECTS

Emotion Recognition System (Python, Pandas, NumPy, TensorFlow, Keras)

- Built a CNN-based model using the FER2013 dataset (35K images), improving accuracy from 64% to 70% through a custom preprocessing pipeline (normalization, augmentation).
- Trained over 50 epochs, reducing validation loss by 18%, and implemented early stopping to prevent overfitting.

Object Recognition System (Python, TensorFlow, Deep Learning)

- Developed an image classification model using the CIFAR-10 dataset (60K images across 10 classes), achieving 85% accuracy after hyperparameter tuning.
- Applied dropout regularization and batch normalization, reducing validation loss by 12%.

Real-Time Object Tracking System (Python, OpenCV)

- Designed a vehicle tracking system using contour detection and background subtraction on video feeds.
- Achieved 92% detection accuracy at 18 FPS on Intel i5, tested on 2-minute traffic footage under daylight conditions.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL

ML & Robotics: Scikit-learn, NumPy, Pandas, TensorFlow, OpenCV, ROS2 (Humble), RTDE, MoveIt, RViz

Databases & DevOps: MySQL, PostgreSQL, Docker, Git, Jira, Red Hat, Ubuntu 22.04

SCHOLARSHIPS & AWARDS

Singapore International Pre-Graduate Award (SIPGA) <i>Agency for Science, Technology and Research (A*STAR)</i> Granted a prestigious research award in AI & Robotics at SIMTech.	<i>Sep. 2024 – Dec. 2024</i> <i>Singapore</i>
STudent Advanced Research Training (START) <i>Joint Institute for Nuclear Research (JINR)</i> Granted a fully funded ML & physics research internship at MLIT.	<i>Aug. 2024 – Sep. 2024</i> <i>Russia</i>
U.S. Study Abroad Program <i>United States Agency for International Development (USAID)</i> Granted a fully funded U.S. exchange semester at the University of Louisville.	<i>Aug. 2023 – Dec. 2023</i> <i>USA</i>
USAID Scholars Activity <i>USAID - The American University in Cairo</i> Granted a fully funded bachelor's scholarship at Cairo University.	<i>Oct. 2020 – Jan. 2025</i> <i>Egypt</i>

TEST SCORES

IELTS Academic (18 Nov 2025) - Overall: 8.0 - Reading: 8.5 Listening: 7.5 Writing: 8.0 Speaking: 8.0
